Demo Client · DEMO-0002
11 November 2026 · v1.0

Field	Value
Document	Methodology for holdings-based portfolio risk analysis
Template version	portfolio_risk 1.0.0 / proquant-core 0.1.0
Engagement	DEMO-0002 · Demo Client
Portfolio	Demo Multi-Asset Fund, as of 2025-12-31
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1. Purpose and scope

The analysis measures the risk of the portfolio's **current holdings**: exposures and concentration, volatility and its sources, sensitivity to systematic factors, potential losses at given confidence levels (VaR, Expected Shortfall), losses under historical and hypothetical stress scenarios, and compliance with the client's risk limits. It also tracks these measures month by month. It does not forecast returns, assess the suitability of investments, or measure liquidity risk unless trading-volume data is in scope.

2. Data

2.1 Inputs

Data set	Size	Fingerprint (SHA-256)
holdings	943 rows	`7cc200fef0cfd107`
prices	99,295 rows	`efd3f25952de6bb3`
benchmark	2,609 rows	`5639df02765c8ca5`
factors	2,609 rows	`0e8b02ed5d06d1d4`

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail.

2.2 Cleaning rules (all logged in `tables/cleaning_log.csv`)

1. Lines without a date or instrument identifier are dropped.
2. Exact duplicate holdings lines are dropped (to be confirmed with the administrator); several distinct lines for the same instrument and date are aggregated (lots, accounts).
3. Market value is reconciled to quantity × price; differences above 1% are flagged.
4. Weights are market value / NAV of each snapshot (cash included).
5. Price gaps up to `max_ffill_days` are forward-filled inside the history; stale price runs are flagged.
6. Missing returns of held instruments (short history, gaps) are filled with the average return of peers in the same asset class and sector (else asset class). Proxied days are counted and reported.

7. Cash positions earn a zero price return in base currency.

3. Exposures and concentration

Exposures are summed weights by asset class, sector, country and currency. Concentration measures exclude cash: largest position, top-10 weight, Herfindahl index HHI = $\Sigma\, w_i^2 / (\Sigma|w_i|)^2$ and effective number of positions 1/HHI.

4. Volatility, correlation and risk contributions

Pro-forma returns apply current weights w to the last `lookback_days` daily instrument returns. The covariance matrix Σ is exponentially weighted (RiskMetrics, zero mean, λ = `ewma_lambda`) or the equally weighted sample covariance.

- Ex-ante volatility $\sigma_p = \sqrt{(w'\Sigma w \cdot 252)}$.
- Euler risk contributions: $RC_i = w_i\,(\Sigma w)_i / \sigma_p$; $\Sigma_i\, RC_i = \sigma_p$ exactly. Shares are RC_i / σ_p.
- Diversification ratio: $\Sigma\, |w_i|\, \sigma_i / \sigma_p$ (1 = no diversification benefit).
- Average pairwise correlation of the 15 largest positions.

5. Factor exposure

Pro-forma daily returns are regressed on factor returns with an intercept (OLS). Standard errors are Newey-West (Bartlett kernel, `newey_west_lags` lags). Systematic variance is $\beta'\Sigma_F\,\beta$ (annualised); the remainder of total variance is idiosyncratic. Instrument-level betas (OLS) are used for hypothetical stress tests.

6. Value-at-Risk and Expected Shortfall

All figures are losses as a fraction of NAV over one day, scaled by √h to h days.

Method	VaR at confidence α	ES at α
Historical simulation	$-\text{quantile}_{1-\alpha}$ of pro-forma daily P&L	−mean of P&L ≤ that quantile
Parametric (normal)	$z_\alpha\,\sigma$	$\sigma\,\phi(z_\alpha) / (1-\alpha)$
Cornish-Fisher	$-(\tilde{z}\,\sigma)$, $\tilde{z}$ adjusted for skewness S and excess kurtosis K	–
Monte Carlo	quantile of simulated P&L w'X, X ~ multivariate normal or Student-t (ν d.o.f.) rescaled to covariance Σ	mean beyond VaR

Cornish-Fisher: $\tilde{z} = z + (z^2-1)S/6 + (z^3-3z)K/24 - (2z^3-5z)S^2/36$, with $z = \Phi^{-1}(1-\alpha)$. Component VaR (parametric) $z_\alpha\, w_i(\Sigma w)_i/\sigma$ sums to total parametric VaR.

6.1 Backtesting

For each day of the realized-return history, historical VaR is computed from the weights held on the previous day and the preceding `window` days of instrument returns, then compared with that day's realized portfolio return. Tests: * **Kupiec proportion of failures:** LR = −2 ln[(1−p)^(n−x) p^x / ((1−x/n)^(n−x) (x/n)^x)] ~ $\chi^2(1)$. * **Christoffersen independence:** do exceptions cluster? LR ~ $\chi^2(1)$. * **Basel traffic light:** GREEN if the binomial probability of ≤ x exceptions is below 95%, YELLOW below 99.99%, RED otherwise.

7. Stress tests

- **Historical:** each instrument's compounded return over the scenario window applied to current weights; instruments without data in the window use peer proxies (the proxied weight is reported). The benchmark's return over the same window is shown for comparison.
- **Hypothetical:** factor shocks s mapped to instruments through their factor betas ($r_i = \beta_i$'s) and aggregated with current weights. Linear approximation.

8. Limits and monitoring

Each limit from the client's risk policy is compared with the measured value. Utilisation = value / limit (or limit / value for minimum limits). Status: BREACH above 100%, WATCH at or above `watch_threshold`, otherwise OK. Overall status: any BREACH → "Limit breach"; any WATCH → "Within limits, items on watch"; otherwise "Within all limits". The same measures are computed at every month-end snapshot (monitoring history) and compared with the previous month.

9. Assumptions, limitations and deviations

Historical estimation (regimes absent from the look-back are not captured); square-root-of-time scaling; peer proxies for short histories; linear hypothetical stresses; no liquidity risk without volume data; data as supplied by the client / administrator. No deviations from the standard methodology.

10. Reproducibility

Each run folder contains `config_used.yaml`, `audit.json` (analyst, timestamps, code version, input fingerprints, output hashes), `run.log`, every result table as CSV and all documents. Monte Carlo uses a fixed seed, so re-running with the same inputs reproduces all results.

11. References

- J.P. Morgan / Reuters (1996). *RiskMetrics Technical Document*, 4th ed.
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- Kupiec, P. (1995). Techniques for Verifying the Accuracy of Risk Measurement Models. *Journal of Derivatives*, 3(2).
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